Lecture Notes III

Course: Analytical Mechanics

Topic: Variational symmetries of Lagrangians

Reference: G. F. Torres del Castillo, *Relating the Lagrange equations with variational principles*, Rev. Mex. Fís. E **24**(1) (2027), in press.

Throughout this chapter, we will let μ be the Lebesgue measure on $\mathbb{R}$ and use the notation

$$\int_{\delta t} \mathrm{d}\mu(t) \equiv \int_{t_0}^{t_1} \mathrm{d}t, \quad \mathbb{R} \supset \delta t := (t_0, t_1).$$

The notes are my way if understanding the article referenced above, I expanded some computations presented there and all the credit for the discussions and equations are for Gerardo F. Torres del Castillo.

The Euler-Lagrange equations revisited

Let us now formally show that Euler-Lagrange equations are the necessary conditions for an integral of the form

$$\int_{\delta t} \mathrm{d}\mu(t)\, L(q_i, \dot{q}_i, t) \tag{$*$}$$

to have an extreme value (in the sense to be explained below). The reason to do such a tedious calculation is to acquire the tools needed to study the variational symmetries of a Lagrangian.

Given any measurable function $L : \mathbb{R}^{2n+1} \to \mathbb{R}$, and a curve $C : [t_0, t_1] \to \mathbb{R}^{n+1}$ defined by $C(t) = (q_i(t), t)$, we define the curve $\overline{C}(t) = (q_i(t), \mathrm{d}q_i(t)/\mathrm{d}t, t)$, and we consider the line integral

$$I(C) = \int_{\overline{C}} \mathrm{d}t\, L, \tag{$\dagger$}$$

which is, by definition, the integral in $(*)$. Now, in order to find the values that extremise the value of I, we shall compare it with the value of I on different "neighbouring" paths, i.e., paths defined by a one-parameter family of transformations:

$$C^{(s)} = \left(q_i^{(s)}, t^{(s)}\right) \quad q_i^{(s)} = q_i^{(s)}(q_j(t), t, s), \quad t^{(s)} = t^{(s)}(q_j(t), t, s), \tag{$**$}$$

labelled by a real parameter s, which can take values in some neighbourhood of zero, this is sometimes understood as s being "small" although the neighbourhood could be "large", but we do not assume anything about the size of s other that being defined on some neighbourhood of s. The transformation should be such that $C^{(0)} = C$, i.e., $q_i^{(0)} = q_i$ and $t^{(0)} = t$. An example of such a transformation is given by

$$x^{(s)} = x\cos s - y\sin s, \quad y^{(s)} = x\sin s + y\cos s, \quad t^{(s)} = e^s t,$$

notice that–in this example–the value s can take arbitrary values on $\mathbb{R}$. The transformations in $(**)$ are considered as *active*[1] transformations. We want to find a condition that a curve C has to satisfy in order for the integral $I(C)$ to have an extreme value, and to do so, we compare it to its values on the neighbouring paths $C^{(s)}$, obtained by displacing the curve C by means of $(**)$. The numerical value of $I(C^{(s)})$ will, in general, depend on s because although the integral has the same definition, the curve may change

$$I(C^{(s)}) = \int_{\overline{C^{(s)}}} \mathrm{d}t^{(s)}\, L.$$

So notice how we can regard $I(C^{(s)})$ as a real-valued function $s \to I(C^{(s)})$ of a real variable, so that we compute its (usual) derivative $\mathrm{d}I(C^{(s)})/\mathrm{d}s$ at $s = 0$, which must be equal to zero if $I(C)$ is an extreme

[1] An active transformation is one in which the $(n+1)$ points are being displaced while the system of coordinates employed is fixed.

value (when compared to the neighbouring paths $C^{(s)}$ such that the endpoints are the same as those of C). Using Leibnitz' rule (differentiation under the integral sign), integration by parts, and the chain rule one gets

$$\left.\frac{\mathrm{d}I(C^{(s)})}{\mathrm{d}s}\right|_{s=0} = \frac{\mathrm{d}}{\mathrm{d}s}\int_{\delta t^{(s)}} \mathrm{d}\mu\big(t^{(s)}\big)\, L\left(q_i^{(s)}(t), \mathrm{d}q_i^{(s)}/\mathrm{d}t^{(s)}, t^{(s)}(t)\right) = \frac{\mathrm{d}}{\mathrm{d}s}\int_{\delta t} \mathrm{d}\mu\big(t\big)\, \frac{\mathrm{d}t^{(s)}}{\mathrm{d}t} L\left(q_i^{(s)}(t), \frac{\mathrm{d}q_i^{(s)}/\mathrm{d}t}{\mathrm{d}t^{(s)}/\mathrm{d}t}, t^{(s)}(t)\right)\Bigg|_{s=0}$$

$$= \int_{\delta t} \mathrm{d}\mu(t) \left[\frac{\mathrm{d}t^{(s)}}{\mathrm{d}t}\left\{\frac{\partial L}{\partial q_i^{(s)}}\frac{\partial q_i^{(s)}}{\partial s} + \frac{\partial L}{\partial\left(\frac{\mathrm{d}q_i^{(s)}/\mathrm{d}t}{\mathrm{d}t^{(s)}/\mathrm{d}t}\right)} \frac{\mathrm{d}t^{(s)}/\mathrm{d}t \frac{\partial \mathrm{d}q_i^{(s)}/\mathrm{d}t}{\partial s} - \mathrm{d}q_i^{(s)}/\mathrm{d}t \frac{\partial \mathrm{d}t^{(s)}/\mathrm{d}t}{\partial s}}{\big(\mathrm{d}t^{(s)}/\mathrm{d}t\big)^2} + \frac{\partial L}{\partial t^{(s)}}\frac{\partial t^{(s)}}{\partial s}\right\} + L\frac{\partial}{\partial s}\frac{\mathrm{d}t^{(s)}}{\mathrm{d}t}\right]_{s=0}$$

$$\implies \left.\frac{\mathrm{d}I(C^{(s)})}{\mathrm{d}s}\right|_{s=0} = \int_{\delta t} \mathrm{d}\mu(t)\left(\frac{\partial L}{\partial q_i}\eta_i + \frac{\partial L}{\partial \dot{q}_i}\big(\dot{\eta}_i - \dot{q}_i\dot{\xi}\big) + \frac{\partial L}{\partial t}\xi + L\dot{\xi}\right) \tag{‡}$$

$$\begin{aligned}
&= \int_{\delta t} \mathrm{d}\mu(t)\left(\frac{\partial L}{\partial q_i}\eta_i + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\eta_i\right) - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\eta_i - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\dot{\xi} + \frac{\partial L}{\partial t}\xi + L\dot{\xi}\right)\\
&= \int_{\delta t} \mathrm{d}\mu(t)\left(\frac{\partial L}{\partial q_i}\eta_i + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\eta_i\right) - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\eta_i + \frac{\partial L}{\partial t}\xi + \dot{\xi}\left(L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right)\right)\\
&= \int_{\delta t} \mathrm{d}\mu(t)\left(\frac{\partial L}{\partial q_i}\eta_i + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\eta_i\right) - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\eta_i + \frac{\partial L}{\partial t}\xi + \frac{\mathrm{d}}{\mathrm{d}t}\left(\xi\left[L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right]\right) - \xi\frac{\mathrm{d}}{\mathrm{d}t}\left(L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right)\right)\\
&= \left[\frac{\partial L}{\partial \dot{q}_i}\eta_i + \xi\left(L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right)\right]_{t_0}^{t_1} + \int_{\delta t} \mathrm{d}\mu(t)\left(\frac{\partial L}{\partial q_i}\eta_i - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\eta_i + \frac{\partial L}{\partial t}\xi - \xi\frac{\mathrm{d}}{\mathrm{d}t}\left(L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right)\right),
\end{aligned}$$

$$\left.\frac{\mathrm{d}I(C^{(s)})}{\mathrm{d}s}\right|_{s=0} = \left[\frac{\partial L}{\partial \dot{q}_i}\eta_i + \xi\left(L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right)\right]_{t_0}^{t_1} + \int_{\delta t} \mathrm{d}\mu(t)\left(\frac{\partial L}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\right)\big(\eta_i - \dot{q}_i\xi\big) \tag{††}$$

where we have used the definitions

$$\delta t^{(s)} := (t_0^{(s)}, t_1^{(s)}), \quad \dot{q}_i := \frac{\mathrm{d}q_i}{\mathrm{d}t}, \quad \eta_i := \left.\frac{\partial q_i^{(s)}}{\partial s}\right|_{s=0}, \quad \xi := \left.\frac{\partial t^{(s)}}{\partial s}\right|_{s=0}, \quad \dot{\eta}_i := \frac{\mathrm{d}\eta}{\mathrm{d}t}, \quad \dot{\xi} := \frac{\mathrm{d}\xi}{\mathrm{d}t},$$

and the fact that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right) = \dot{q}_i\left(\frac{\partial L}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\right) + \frac{\partial L}{\partial t}.$$

As we said before, we have the requirement that $\mathrm{d}I(C^{(s)})/\mathrm{d}s = 0$ at $s = 0$ provided that

$$\eta_i(C(t_0)) = \eta_i(C(t_1)) = 0 \quad \text{and} \quad \xi(C(t_0)) = \xi(C(t_1)) = 0 \tag{***}$$

where the latter amounts to saying that the endpoints of C are not displaced in $C^{(s)}$ at $s = 0$. Hence, under such conditions, (††) reduces to

$$\int_{\delta t} \mathrm{d}\mu(t)\left(\frac{\partial L}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\right)\big(\eta_i - \dot{q}_i\xi\big)$$

which must hold for all ξ, η_i satisfying (***), and therefore, by appealing to the DuBois-Reymond Lemma, one gets that

$$\frac{\partial L}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} = 0$$

must hold for each $i = 1, \ldots, n$, which are known as the *Euler-Lagrange equations*.

Time-independent Lagrangians. Local time. The least action principle

We shall now assume that the Lagrangian L satisfies $\partial L/\partial t = 0$. Then $J = p_i\dot{q}_i - L$ is a constant of the motion as we proved earlier in the course. Assuming that C satisfies the Euler-Lagrange equations and denoting by E the constant value of J (which is not always the energy, but we choose the letter E nonetheless) on the curve $\overline{C}$, according to the definition of ξ, we can express the boundary term on the right-hand side of (††) in the form

$$-E\xi\Big|_{t_0}^{t_1} = -E\int_{\delta t}\mathrm{d}\mu(t)\,\dot{\xi} = -E\int_{\delta t}\mathrm{d}\mu(t)\,\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial t^{(s)}}{\partial s}\bigg|_{s=0} = -E\frac{\mathrm{d}}{\mathrm{d}s}\int_{\delta t}\mathrm{d}\mu(t)\frac{\mathrm{d}t^{(s)}(t)}{\mathrm{d}t}\bigg|_{s=0} = \frac{\mathrm{d}}{\mathrm{d}s}\int_{\overline{C^{(s)}}}(-E)\bigg|_{s=0},$$

which can be incorporated in the left-hand side of (††), giving

$$\frac{\mathrm{d}}{\mathrm{d}s}\int_{\overline{C^{(s)}}}(L+E)\bigg|_{s=0} = \int_{\delta t}\left(\frac{\partial L}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial\dot{q}_i}\right)\left(\eta_i - \dot{q}_i\xi\right) + \left[\frac{\partial L}{\partial\dot{q}_i}\eta_i\right]_{t_0}^{t_1},$$

which shows that if C satisfies the Euler-Lagrange equations, then the integral

$$I_E(C) := \int_{\overline{C}}\mathrm{d}t\,(L+E) \tag{1}$$

has a stationary value on the curve C in comparison with the curves with the same values of $q_1,\ldots,q_n$ at the endpoints as C, without restriction on the values of t at the endpoints. In the application of this result it is convenient to assume that J has a constant value E in all curves being compared. Then with the aid of the equation $J = E$ one can eliminate from I_E all reference to t. Since in (1) the coordinate t does not appear in L, in order to calculate the integral $I_E(C)$ we only have to give the coordinates q_i as functions of the parameter to be employed, but it is not necessary to give t as a function of that parameter; this allows to introduce a new variable, say τ, such that

$$\mathrm{d}t = f(q_i)\mathrm{d}\tau \tag{2}$$

where $f(q_i)$ is any nonnegative (to keep the orientation of the integral) function of the coordinates q_i only. Then

$$I_E(C) = \int_{\delta\tau}\mathrm{d}\mu(\tau)\,(L+E)f \equiv \int_{\delta\tau}\mathrm{d}\mu(\tau)\,\widetilde{L}_E$$

where $\delta\tau := (\tau_0,\tau_1)$ and τ_0,τ_1 are the (here, irrelevant to know) values of τ at the endpoints of C and $\widetilde{L}_E = (L+E)f$. We can easily convince ourselves that

$$\frac{\mathrm{d}}{\mathrm{d}s}I_E(C^{(s)})\bigg|_{s=0} = \int_{\delta\tau}\mathrm{d}\mu(\tau)\left(\frac{\partial\widetilde{L}_E}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial\widetilde{L}_E}{\partial q_i'}\right)\left(\eta_i - q_i'\xi\right) + \left[\frac{\partial\widetilde{L}_E}{\partial q_i'}\eta_i - \xi\left(\widetilde{L}_E - \frac{\partial\widetilde{L}_E}{\partial\dot{q}_i}\dot{q}_i\right)\right]_{\tau_0}^{\tau_1},$$

where $q_i' \equiv \mathrm{d}q_i/\mathrm{d}\tau$, and since

$$\frac{\partial\widetilde{L}_E}{\partial q_i'}q_i' - \widetilde{L}_E = \frac{\partial(L+E)f}{\partial\dot{q}_i}\dot{q}_i - (L+E)f = f\left(\frac{\partial L}{\partial\dot{q}_i}\dot{q}_i - L - E\right) = f(J-E)$$

we get that the Jacobi integral corresponding to $\widetilde{L}_E$ at the endpoints of $\overline{C}$ is zero so

$$\frac{\mathrm{d}}{\mathrm{d}s}I_E(C^{(s)})\bigg|_{s=0} = \int_{\delta\tau}\mathrm{d}\mu(\tau)\left(\frac{\partial\widetilde{L}_E}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial\widetilde{L}_E}{\partial q_i'}\right)\left(\eta_i - q_i'\xi\right) + \underbrace{\left[\frac{\partial\widetilde{L}_E}{\partial q_i'}\eta_i\right]_{\tau_0}^{\tau_1}}_{=0},$$

since the endpoints of C must be fixed in the transformation, and hence, again, the boundary term vanishes, this shows that the Euler-Lagrange equations are equivalent to

$$\frac{\partial\widetilde{L}_E}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial\widetilde{L}_E}{\partial q_i'} = 0. \tag{3}$$

It may be remarked that the freedom of f gives us the option to look for such an f such that (3) are simpler than the original Euler-Lagrange equations. Owing to its role, the parameter τ is called **fictitious time** (the name arising from the fact that τ may not have units of time) or **local time**. It is important to emphasise that, by contrast with $(**)$ which gives $t^{(s)}$ in terms of q_i, t, the correspondence $t(\tau)$ would be obtained from (2) once we've got the coordinates q_i in terms of τ.

Taking the definition of $I_E(C)$ one can write such an integral in the form

$$I_E(C) = \int_{\overline{C}} \mathrm{d}t \left(\frac{\partial L}{\partial \dot{q}_i} - J + E \right) = \int_{\overline{C}} \mathrm{d}t \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i = \int_{\overline{C}} \mathrm{d}q_i p_i,$$

this integral is called the **action** and the result proved above is the *least action principle* (formulated by Euler in 1744, following the initial ideas of Maupertius).

Variational symmetries of Lagrangians

As we mentioned earlier, the procedure followed here will allow us to deduce the basic formulas related to variational symmetries. We begin by giving the basic definition of this section: the one-parameter family of transformations $(**)$ is said to be a family of **strict symmetries** of a Lagrangian L if

$$I(C^{(s)}) - I(C) = 0 \tag{i}$$

for any curve C and all possible values s. Then it is clear that

$$\left. \frac{\mathrm{d}I(C^{(s)})}{\mathrm{d}s} \right|_{s=0} = 0$$

and thus one can readily verify that for a curve C satisfying the Euler-Lagrange equations one gets

$$\left[\frac{\partial L}{\partial \dot{q}_i} \eta_i + \xi \left(L - \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right) \right]_{t_0}^{t_1} = 0$$

for all t_0, t_1, which means that the quantity inside the brackets is an integral of the motion, and by aid of the definition of J we have that

$$\frac{\partial L}{\partial \dot{q}_i} \eta_i - J\xi \tag{ii}$$

is the integral of the motion, and also, using $(‡)$ and under the same assumptions mentioned before, we get that

$$\left. \frac{\mathrm{d}}{\mathrm{d}s} I_E(C^{(s)}) \right|_{s=0} = \int_{\delta t} \mathrm{d}\mu(t) \left(\frac{\partial L}{\partial q_i} \eta_i + \frac{\partial L}{\partial \dot{q}_i} \left(\dot{\eta}_i - \dot{q}_i \dot{\xi} \right) + \frac{\partial L}{\partial t} \xi + L\dot{\xi} \right) = 0$$

so that the arbitrariness of $[t_0, t_1]$ yields

$$\frac{\partial L}{\partial q_i} \eta_i + \frac{\partial L}{\partial \dot{q}_i} \left(\dot{\eta}_i - \dot{q}_i \dot{\xi} \right) + \frac{\partial L}{\partial t} \xi + L\dot{\xi} = 0$$

or, equivalently,

$$\frac{\partial L}{\partial q_i} \eta_i + \frac{\partial L}{\partial \dot{q}_i} \dot{\eta}_i + \frac{\partial L}{\partial t} \xi + \dot{\xi} \left(L - \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right) = 0$$

the latter equation determines the one-parameter families of strict symmetries of L and, once we find the functions η_i and ξ, we get "for free" that the quantity in (ii) is an integral of the motion. In this sense, η_i, ξ are called *infinitesimal transformations*. The concept of a strict symmetry of a Lagrangian can be conveniently relaxed by replacing the condition (i) by the weaker one

$$I(C^{(s)}) - I(C) = F(C(t_1), s) - F(C(t_0), s) \tag{iii}$$

where F is some function of the coordinates q_i, the time t and the parameter s only. The transformations satisfying (iii) are usually called **variational symmetries** but are also known as **Noether-Bessel-Hagen symmetries**. From (iii) we obtain

$$\left. \frac{\mathrm{d}I(C^{(s)})}{\mathrm{d}s} \right|_{s=0} = G(C(t_1)) - G(C(t_0)) = \int_{\delta t} \mathrm{d}\mu(t) \, \frac{\mathrm{d}G}{\mathrm{d}t}, \quad G := \left. \frac{\partial F}{\partial s} \right|_{s=0},$$

which substituted back into (‡) yields

$$\frac{\partial L}{\partial q_i}\eta_i + \frac{\partial L}{\partial \dot{q}_i}\left(\dot{\eta}_i - \dot{q}_i\dot{\xi}\right) + \frac{\partial L}{\partial t}\xi + L\dot{\xi} = \frac{\mathrm{d}G}{\mathrm{d}t},$$

or, equivalently,

$$\frac{\partial L}{\partial q_i}\eta_i + \frac{\partial L}{\partial \dot{q}_i}\frac{\mathrm{d}\eta_i}{\mathrm{d}t} + \frac{\partial L}{\partial t}\xi + \frac{\mathrm{d}\xi}{\mathrm{d}t}\left(L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right) = \frac{\mathrm{d}G}{\mathrm{d}t}, \tag{iv}$$

which by (††) gives that

$$N \equiv \frac{\partial L}{\partial \dot{q}_i}\eta_i - J\xi - G$$

is an integral of the motion. We will consider (iv) as the most general condition for symmetries in a Lagrangian, and note that here, for G constant, we get back the condition for a strict symmetry. The inclusion of G can be very useful since there are Lagrangians that do not have a family of strict symmetries, but do have nontrivial variational symmetries.

Worked-out examples

Relation to previous results: We can readily see that the results of the previous section reduce to the results presented in the earlier sessions of the course for some choices of η_i, ξ.

In particular, if one has that q_k is ignorable, i.e., $\partial L/\partial q_k = 0$ for some $1 \le k \le n$, then (iv) are satisfied with $\eta_i = \delta_{ik}, \xi = 0, G = 0$ and this particular choice yields the integral

$$N = \frac{\partial L}{\partial \dot{q}_k}.$$

Another example of a previous symmetry is given by taking $\eta_i = \delta_{ik}, \xi = 0$ which turns (iv) into

$$\frac{\partial L}{\partial q_k} = \frac{\mathrm{d}G}{\mathrm{d}t} = \frac{\partial G}{\partial q_i}\dot{q}_i + \frac{\partial G}{\partial t}$$

since G is a function of the coordinates, time and possibly s itself (s is an independent parameter, so it does not appear in the time derivative of G). The previous equation was deduced earlier by inspection. The associated integral (as we saw earlier) is given by

$$N = \frac{\partial L}{\partial \dot{q}_k} - G.$$

For the next examples, we will use the formula

$$I(C^{(s)}) = \int_{\delta t} \mathrm{d}\mu(t)\, \frac{\mathrm{d}t^{(s)}}{\mathrm{d}t} L\left(q_i^{(s)}(t), \frac{\mathrm{d}q_i^{(s)}/\mathrm{d}t}{\mathrm{d}t^{(s)}/\mathrm{d}t}, t^{(s)}(t)\right).$$

Example 1: Let us consider the Lagrangian

$$L = \frac{1}{2}t\dot{q}^2 - ate^q$$

which gives rise to the Poisson-Boltzmann equation. We also consider the transformations

$$t^{(s)} = e^{cs}t, \quad q^{(s)} = q - 2cs,$$

then

$$\frac{\mathrm{d}t^{(s)}}{\mathrm{d}t} = e^{cs}, \quad \frac{\mathrm{d}q^{(s)}}{\mathrm{d}t} = \frac{\mathrm{d}q}{\mathrm{d}t} \implies \frac{\mathrm{d}q^{(s)}/\mathrm{d}t}{\mathrm{d}t^{(s)}/\mathrm{d}t} = e^{-cs}\frac{\mathrm{d}q}{\mathrm{d}t}$$

$$I(C^{(s)}) = \int_{\delta t} \mathrm{d}\mu(t) e^{cs} e^{cs} t\left(\frac{1}{2}e^{-2cs}\dot{q}^2 - ae^{q-2cs}\right) = I(C)$$

so that the transformations leave strictly invariant the Lagrangian. Next, we compute

$$\eta = \left.\frac{\partial q^{(s)}}{\partial s}\right|_{s=0} = -2c, \quad \xi = \left.\frac{\partial t^{(s)}}{\partial s}\right|_{s=0} = ct$$

so that the integral of the motion is

$$N = \frac{\partial L}{\partial \dot{q}}\eta - J\xi = t\dot{q}(-2c) - \left(\frac{1}{2}t\dot{q}^2 + ate^q\right)ct = ct\left(-2\dot{q} - \frac{1}{2}t\dot{q}^2 + ate^q\right).$$

Example 2: Consider now the Lagrangian $L = \frac{1}{2}m\dot{q}^2 - \alpha/q^2$ with the transformations

$$q^{(s)} = \frac{q}{1-st}, \quad t^{(s)} = \frac{t}{1-st},$$

then

$$\frac{\mathrm{d}t^{(s)}}{\mathrm{d}t} = \frac{1}{(1-st)^2}, \quad \frac{\mathrm{d}q_i^{(s)}}{\mathrm{d}t} = \frac{qs}{(1-st)^2},$$

so that

$$I(C^{(s)}) = \int_{\delta t} \mathrm{d}\mu(t)\frac{1}{(1-st)^2}\left(\frac{1}{2}m[(1-st)\dot{q} + qs]^2 - \frac{\alpha}{q^2}(1-st)^2\right)$$

and

$$I(C^{(s)}) - I(C) = \frac{1}{2}m\int_{\delta t}\mathrm{d}\mu(t)\left(\frac{q^2s^2}{(1-st)^2} + \frac{2q\dot{q}s}{(1-st)}\right) = \int_{\delta t}\mathrm{d}\mu(t)\frac{\mathrm{d}}{\mathrm{d}t}\left(q^2\frac{s}{1-st}\right)$$

where we used the fact that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{s}{1-st}\right) = \frac{s^2}{(1-st)^2}$$

and integration by parts. We therefore conclude that

$$I(C^{(s)}) - I(C) = \frac{1}{2}m\left[q^2\frac{s}{1-st}\right]_{t_0}^{t_1},$$

which clearly suggests (cf. equation iii) that

$$F(q,t,s) = \frac{1}{2}mq^2\frac{s}{1-st},$$

whence

$$G \equiv \left.\frac{\partial F}{\partial s}\right|_{s=0} = \frac{1}{2}mq^2,$$

and thus the integral of the motion is just

$$N = \left(\frac{1}{2}m\dot{q}^2 + \frac{\alpha}{q^2}\right)t^2 - mq\dot{q}t + \frac{1}{2}mq^2.$$